TIM 172A, MOT-I, Lecture #2, 9/30/25

Agenda :

1. Quick summary of L#1

2. HW #1 (due on Thursday, 10/2/25)

3. TIM 172P
   - placed all students in group
   - posted the Project Proposal assignment
   - for the whole TIM 172A, the "FILES" tab on Canvas will contain a "TIM 172P" folder

TIM 172A IMPORTANT NOTE:

- All submitted work must be TYPEWRITTEN
- We will NOT accept handwritten work
- Figures, graphics can be hand-drawn, & inserted in the typewritten text.

1. Quick summary of L#1

- Fast cycle competitors - FAANG & MANGA — can balance the conflicting trade-offs between product quality, speed, & cost; and can successfully manage risk.
- This course will develop the building blocks, & a framework - MDC - for integrating these blocks
- Professionals (Engineering, Business, ....) generally use a 5-step structured problem-solving process (SPSP):

Define → Plan (or Process or Approach) → Execute → Checking → Learn/Generalize

2. HW # 1, Prob. # 1

Pre-liminary instructions (for all the work – homeworks, exams, project tasks – done in TIM 172A/P and beyond.):

1. Develop a planning habit
   – Create a simple plan (what, when, how long, .....) for doing all the problems in an assigment

2. Develop a structured problem-solving habit.
   – Download the SPSP handout on Canvas
   – For every problem, "copy-and-paste" the handout into your proposed solution
   – Decide which sub-steps (for each of the 5 primary steps) are relevant or necessary to the solution of your particular problem
     ⇒ Customize the SPSP for your particular problem.

(back to) HW # 1, Prob # 1

Problem as given : "Improve the existing cell-phone."

$\Rightarrow$ Vague, open-ended statement

Apply the SPSP to this problem

Step 1 : Define the (real) problem (requires reflection, thinking, asking questions)

Rough conceptual sketch } $\boxed{\text{Customer Needs}} \rightarrow \boxed{\text{Assess} \cdots\cdots} \rightarrow \boxed{\text{Improve}}$

Establish the sub-problems (SPs)

SP1 : Establish customer needs that all (generic) cell-phones should satisfy

SP2 : Assess existing cell-phone products to determine how well these customer needs are satisfied

w.r.t. : with respect to

SP3 : Identify (generate) ways (or solutions) to improve existing cell-phones w.r.t. (a) customer needs (from SP1) & assessment (from SP2)

## Step 2 : Create a PLAN

(a) Assumptions

whom am I ? : perhaps an engineer or a marketing analyst at a cell-phone company like Apple, Samsung, ...

who is the solution for : for the new product development team, ...., CTO, CEO

which type (manufacturer) of cell-phones should I (you) focus on : OBVIOUS ("Be practical")

(b) What information do I need ?

- how cell-phones work ⇒ functions & customer needs of cell-phones, & how they are realized (made real) or FORM

("HOWSTUFFWORKS.COM")

- other (?)

Create a plan for solving each sub-problem (SP)

SP1 : Establish needs .......

Plan
- how cell-phone work
- Internet research
- list of customer needs (functions, features, ...)
- ?
- ?

SP2 : Assess how well ....
- your own experience
- internet consumer research
- customer surveys, ...
- ...
- ...

SP3 : Improve
- research (internet)
- creative brainstorming
- ... ?

## Step 3 : Execute the plan

of the SPSP process

SP1 : $\underbrace{\text{(from the plan)}}_{\text{steps}}$ : execute plan : 1-2 hrs

SP2 : . . . . . execute plan : 2-4hrs

SP3 : . . . . . . execute plan : 2-4 hrs

How should I present my results :

Table ?

| | Needs | Assessment | ? | ? |
|---|---|---|---|---|
| examples | ease of use | | | |
| | performance | | | |
| | reliability | | | |
| | . . .<br>. . . | | | |

Step 4 (of the SPSP) : Check your work
( Reflect, ... ask questions, ... )

Step 5 : Learn & generalize
- about cell-phones & improving them
- yourself as a problem-solver

HW # 1 , Prob 2

simpler version of Prob 1

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TA structure for TIM 172A

Infrastructure (Canvas, .... ) — Vihan

Customer & Grader facing — Fariha

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